

Samrudha Narendra

Bengaluru, India

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EDUCATION

PES University

B.Tech in Computer Science Engineering — CGPA: 8.28/10

2023 – 2027

Bengaluru, India

National Public School, Rajajinagar

Senior High School — Percentage: 91.2%

2019 – 2023

EXPERIENCE

Summer Research Intern

CCBD-CDASML Research Centre, PES University

June 2025 – July 2025

Bengaluru, India

- Fine-tuned open-source Whisper LLM on Kannada speech corpora using PyTorch with phoneme-level annotations, optimizing model performance through dataset curation and alignment constraint analysis
- Built evaluation pipelines measuring WER/CER across gender-stratified datasets; prepared and cleaned training datasets for ML model optimization
- Integrated fine-tuned ASR model into an inference pipeline, simulating production-grade AI model deployment workflow
- Published peer-reviewed research at IEEE I3CTCON 2026 [\[Paper\]](#)

Teaching Assistant – C Programming

PES University

Feb 2026 – June 2026

Bengaluru, India

- Mentored 65+ undergraduates across weekly lab sessions in C programming, memory management, pointers, and data structures; graded 100+ assignments per semester and resolved recurring coding issues, improving concept retention and reducing average debugging time

PROJECTS

Hybrid AI Product Search Engine | Python, FastAPI, Node.js, FAISS, BM25, LightGBM, Kafka, MongoDB, Docker

- Architected a production-grade search backend with BM25 + FAISS retrieval and LightGBM LambdaRank re-ranking; exposed via secured REST APIs with sub-second latency
- Implemented Reciprocal Rank Fusion (RRF) with real-time filtering on category, price, and inventory constraints using a Kafka-driven microservices architecture deployed via Docker
- Designed scalable backend system architecture supporting real-time product indexing and horizontal scaling – built for real users, not demo use

Self-Healing Agentic RAG System | Python, LangChain, FAISS, BM25, HuggingFace, Vector DB

- Built and deployed a production-ready RAG pipeline with hybrid BM25 + FAISS vector retrieval, automated hallucination detection, query reformulation, and self-correction workflows validated on HotpotQA

RideSync – Campus Ride-Sharing Platform | Java, Spring Boot, JWT, Spring Security, REST APIs, MySQL, Hibernate

- Secured 40+ REST APIs using JWT authentication and Spring Security with role-based access control; implemented OTP verification, banned-user enforcement, and full audit logging
- Built production-ready backend with Dijkstra route optimization, JPA repositories, ride history, and ratings in a layered MVC architecture

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, SQL

Frameworks & Libraries: Node.js, Express.js, React.js, Spring Boot, FastAPI, PyTorch, LangChain, HuggingFace

AI / ML: LLMs, RAG, FAISS, BM25, Vector Databases, Whisper, Transformers, Model Fine-tuning, Dataset Preparation

Databases: MongoDB, MySQL, PostgreSQL, Redis

Cloud & DevOps: Docker, Kafka, Git, GitHub, CI/CD, AWS, REST APIs, Linux

Core CS: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP

Club Head – GronIT Club, PES University

2025 – 2026

- Spearheaded ideathons and a state-level hackathon with 100+ participants; managed sponsorships, outreach, logistics, and a 60-member organizing team end-to-end
- Drove club growth through strategic partnerships and cross-campus outreach, expanding event reach and participant engagement year-over-year

Event Management Core – Aikya Club, PES University

2025 – 2026

- Planned and executed 5+ community and technical events; coordinated cross-functional teams of 10–20 members across logistics, venue, and operations